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From yesterday

Reduce Resolvent

$$R_Q(\omega) = \sum_{m>1} \frac{|\Phi_m\rangle\langle\Phi_m|}{\omega - \varepsilon_m}$$

$$\hat{H}_0 |\Phi_m\rangle = \varepsilon_m |\Phi_m\rangle$$

$$\hat{h}_0 |q_i\rangle = \varepsilon_i |q_i\rangle$$

The state for the ground state

$$|\Psi_0\rangle = \sum_{n=0}^{\infty} \left[R_Q(\omega)(\omega - E_0 + H_1) \right]^n |\Phi_0\rangle$$

$$\hat{P} = |\Phi_0\rangle\langle\Phi_0| \quad \wedge \quad \hat{Q} = \sum_{m>1} |\Phi_m\rangle\langle\Phi_m|$$

$$[\hat{P}, \hat{Q}] = 0 \quad \wedge \quad \hat{P}^2 = \hat{P} \quad \wedge \quad \hat{Q}^2 = \hat{Q}$$

$$\Delta E = \Delta E(\text{MBPT})$$

$$= \sum_{n=0}^{\infty} \langle \Phi_0 | \hat{H}_1 [\hat{R}_Q(\omega)(\omega - E_0 + \hat{H}_1)]^n | \Phi_0 \rangle$$

(i) BWPT : $\omega = E_0$

$$E_0 = \langle \Phi_0 | \hat{H}_0 | \Phi_0 \rangle + \Delta E$$

$$\Delta E = \sum_{i=1}^{\infty} \Delta E^{(i)}$$

First order term $n = 0$

$$\Delta E^{(1)} = \langle \Phi_0 | H_1 | \Phi_0 \rangle$$

2nd-order $n = 1$

$$\Delta E^{(2)} = \sum_{m \geq 1} \langle \Phi_0 | H_1 | \Phi_m \rangle \times \frac{\langle \Phi_m | H_1 | \Phi_0 \rangle}{E_0 - E_m}$$

3rd-order

$$\Delta E^{(3)} = \langle \Phi_0 | H_1 \sum_{m \geq 1} \frac{|\Phi_m\rangle\langle\Phi_m|}{E_0 - E_m} \times H_1 \sum_{m \geq 1} \frac{|\Phi_m\rangle\langle\Phi_m| H_1 |\Phi_0\rangle}{E_0 - E_m}$$

$$R_Q(\omega = E_0) = R_E = \sum_{m \geq 1} \frac{|\Phi_m\rangle\langle\Phi_m|}{E_0 - E_m}$$

$$\Delta E^{(3)} = \langle \Phi_0 | H_1 R_E H_1 R_E H_1 | \Phi_0 \rangle$$

$$= \langle H_1 R_E H_1 R_E H_1 \rangle$$

$$\Delta E^{(1)} = \langle H_1 \rangle$$

$$\Delta E^{(2)} = \langle H_1 R_E H_1 \rangle$$

$$\Delta E^{(3)} = \langle H_1 R_E H_1 R_E H_1 \rangle$$

$$\Delta E^{(4)} = \langle H_1 R_E H_1 R_E H_1 R_E H_1 \rangle$$

write it as a finite expression

A is invertible, A − B is also invertible

$$(A-B)^{-1} = A^{-1} + A^{-1}B(A-B)^{-1}$$

Proof: multiply with A − B from the right

$$\mathbb{1} = A^{-1}(A-B) + A^{-1}B\,\mathbb{1}$$

$$= A^{-1}A = \mathbb{1}$$

Def: $A = E_0 - QH_0Q$

$$H_0Q = \sum_{m>1} H_0 |\Phi_m\rangle\langle\Phi_m|$$

$B = QH_1Q$

Iterate $(A-B)^{-1}$

$$A^{-1} + A^{-1}BA^{-1} + A^{-1}BA^{-1}BA^{-1} + \ldots$$

$$A^{-1} = (E_0 - QH_0Q)^{-1} = R_E$$

$$B = QH_1Q$$

$$\cdot (E_0 - QH_0Q)^{-1}$$

$$+ (E_0 - QH_0Q)^{-1} H_1 (E_0 - QH_0Q)^{-1}$$

$$+ (E_0 - QH_0Q)^{-1} H_1 (E_0 - QH_0Q)^{-1} H_1$$

$$\times (E_0 - QH_0Q)^{-1} + \ldots$$

$$= R_E + R_E H_1 R_E + R_E H_1 R_E H_1 R_E + \ldots$$

$$= \left(E_0 - Q H_0 Q - Q H_1 Q\right)^{-1}$$

$\Rightarrow$ BWPT:

$$\langle \Phi_0 | H_1 | \Phi_0 \rangle +$$

$$\langle \Phi_0 | H_1 \left(E_0 - Q H_0 Q - Q H_1 Q\right)^{-1} H_1 | \Phi_0 \rangle$$

$$= \Delta E \text{ (BWPT)}$$

Back to simple example

$|1\rangle = a_1^\dagger |0\rangle \qquad H_0 |1\rangle = \varepsilon_1 |1\rangle$

$H_0 |2\rangle = \varepsilon_2 |2\rangle \qquad \varepsilon_2 > \varepsilon_1$

$$\langle 1|H_1|1\rangle = g = \langle 1|H_1|2\rangle$$

$$= \langle 2|H_1|2\rangle$$

$$H = \begin{bmatrix} \varepsilon_1 + g & g \\ g & \varepsilon_2 + g \end{bmatrix}$$

$$\hat{P} = |1\rangle\langle 1| \quad , \quad \hat{Q} = |2\rangle\langle 2|$$

2 nd - order :

$$\sum_{m \neq 1} \frac{\langle \Phi_0 | H_1 | \Phi_m \rangle \langle \Phi_m | H_1 | \Phi_0 \rangle}{E_0 - \varepsilon_m}$$

$$|\Phi_0\rangle = |1\rangle \qquad = \qquad g^2 / E_0 - \varepsilon_2$$

$$|\Phi_1\rangle = |2\rangle \qquad \varepsilon_0 = \varepsilon_1 \qquad \varepsilon_m = \varepsilon_2$$

Full equation

$$\langle 1|H_1|1\rangle = g = \Delta E^{(1)}$$

$$\underbrace{g}\ \langle 1|H_1|2\rangle\langle 2|\left(E_0 - \underbrace{QH_0Q - QH_1Q}_{(\varepsilon_2 + g)}\right)^{-1}|2\rangle \underbrace{\langle 2|H_1|1\rangle}_{g}$$

$$= \frac{g^2}{E_0 - \varepsilon_2 - g}$$

$$E_0 = \underbrace{\varepsilon_1 + g}_{E^{Ref}} + \frac{g^2}{E_0 - \varepsilon_2 - g}$$

$$\langle 1|H_0|1\rangle \quad \langle 1|H_1|1\rangle$$

$$(E_0 - \varepsilon_1 - g)(E_0 - \varepsilon_2 - g) - g^2 = 0$$

$$= \det(H - E_0)$$

$$= \begin{vmatrix} \varepsilon_1 + g - E_0 & g \\ g & \varepsilon_2 + g - E_0 \end{vmatrix} = 0$$

(ii) Rayleigh-Schrödinger PT (RSPT): $\omega = E_0$

(starting energy)

$$\Delta E = \sum_{n=0}^{\infty} \langle \Phi_0 | H_1 [R_0 (H_1 - \Delta E)]^n | \Phi_0 \rangle$$

$$R_0 = Q (E_0 - H_0)^{-1} Q$$

$$= \frac{Q}{E_0 - H_0} \qquad Q = P_0$$

$$\Delta E_0 = E_0 - \mathcal{E}_0$$

$$\Delta E = \langle \Phi_0 | H_1 | \Phi_0 \rangle$$

$n = 0$

$\Delta E^{(1)}$

$$+ \langle \Phi_0 | H_1 R_0 (H_1 - \Delta E) | \Phi_0 \rangle$$

$n = 1$; $\Delta E^{(2)}$ + . . .

Look at

$$\langle \Phi_0 | H_1 R_0 (H_1 - \Delta E) | \Phi_0 \rangle$$

$$- \langle \Phi_0 | H_1 R_0 \Delta E | \Phi_0 \rangle$$

$$R_0 | \Phi_0 \rangle = \sum_{m \neq 1} \frac{| \Phi_m \rangle \langle \Phi_m | \Phi_0 \rangle}{E_0 - E_m}$$

$$= 0$$

$$\langle \Phi_m | \Phi_0 \rangle$$

$$\Delta E^{(2)} = \langle \Phi_0 | H_1 R_0 H_1 | \Phi_0 \rangle = 0$$

$$\Delta E^{(2)} = \sum_{m>1} \frac{|\langle \Phi_0 | H_1 | \Phi_m \rangle|^2}{\mathcal{E}_0 - \mathcal{E}_m}$$

$$\Delta E^{(3)} = \langle H_1 R_0 (H_1 - \Delta E) R_0 H_1 \rangle$$

$$E_0' - \mathcal{E}_0$$

$$\Delta E = \sum_{n=1}^{\infty} \Delta E^{(n)}$$

$$\Delta E^{(3)} = \langle H_1 R_0 (H_1 - \langle H_1 \rangle) R_0 H_1 \rangle$$

$\Delta E^{(1)}$

$$= \langle H_1 R_0 H_1 R_0 H_1 \rangle - \langle H_1 \rangle \langle H_1 R_0^2 H_1 \rangle$$

$3/(\mathcal{E}_1 - \mathcal{E}_2)^2$

normalization term

Back to simple example

$\langle 1|H_1|1\rangle = g = \langle 2|H_1|2\rangle =$

$\langle 1|H_1|2\rangle$

$\langle 1|H_0|1\rangle = \varepsilon_1 \quad \langle 2|H_0|2\rangle = \varepsilon_2$

$$R_0 = \frac{|2\rangle\langle 2|}{\varepsilon_1 - \varepsilon_2}$$

$$\Delta E = \langle H_1\rangle + \langle H_1 R_0 H_1\rangle$$

$$\langle H_1\rangle = \langle 1|H_1|1\rangle = \langle \Phi_0|H_1|\Phi_0\rangle = g$$

$$\langle H_1 R_0 H_1\rangle = \langle 1|H_1 \frac{|2\rangle\langle 2|}{\varepsilon_1 - \varepsilon_2} H_1|1\rangle = \frac{g^2}{\varepsilon_1 - \varepsilon_2}$$

$$E_0 \simeq \varepsilon_1 + g + \frac{g^2}{\varepsilon_1 - \varepsilon_2}$$ to 2nd order

$$\lambda_{1,2} = \frac{\varepsilon_1 + \varepsilon_2}{2} + g \pm \sqrt{\frac{(\varepsilon_1 - \varepsilon_2)^2}{4} + g^2}$$

$$\lambda_1 = E_0 = \varepsilon_1 + g + \frac{g^2}{\varepsilon_1 - \varepsilon_2} + \cdots$$

$$\Delta E^{(1)} = \langle \Phi_0 | H_1 | \Phi_0 \rangle$$

$$= \frac{1}{2} \sum_{ij} \underbrace{\langle ij | v | ij \rangle_{AS}}_{v_{ijij}}$$

$$H = E_0^{Ref} + F_N + V_N$$

$$F_N = \sum_{pq} \langle p|f|q\rangle a_p^\dagger a_q$$

$$= \sum_{pq} f_{pq}\, a_p^\dagger a_q$$

$$f_{pq} = \langle p|h_0|q\rangle + \sum_j \langle pj|v|qj\rangle_{AS}$$

$$= h_0^{pq} + \sum_j \bar{v}_{pjqj}$$

$$V_N = \frac{1}{4} \sum_{pqrs} \bar{v}_{pqrs}\, a_p^\dagger a_q^\dagger a_s a_r$$

$$E_0^{Ref} = \sum_i h_0^{ii} + \frac{1}{2}\sum_{ij} \bar{v}_{ijij}$$

$$H_1 = E^{(1)} + U_N + V_N$$

$$\langle \Phi_c | H_1 | \Phi_c \rangle$$

$$U_N = \sum_{pq} u_{pq} q_p^+ q_q$$

$$u_{pq} = \sum_{j \le F} \bar{v}_{pj'qj'}$$

Due to Condon-Slater we can have

$$\langle \Phi_c | H_1 | \Phi_m \rangle$$

at 1p1h and 2p2h states as $|\Phi_m\rangle$

$$|\text{PH}\rangle \quad |\Phi_a^q\rangle = a_a^+ q_i \, |\Phi_0\rangle$$

$$\Delta E^{(2)} = \sum_{m \geq 1} \frac{|\langle \Phi_0 | H_1 | \Phi_m \rangle|^2}{E_0 - E_m}$$

$$\langle \Phi_0 | u_W | \Phi_m \rangle$$

$$\sum_{pq} \langle \Phi_0 | u_{pq} \, a_p^+ q_q \, a_a^+ q_i \, | \Phi_0 \rangle$$

$$= u_{ia} \, \delta_{pi} \, \delta_{qa}$$

$$\sum_m |\Phi_m\rangle_{\text{PH}} = \sum_{qi} c_i^a \, q_q^+ q_i \, |\Phi_0\rangle$$

1p1h - term

$$\Delta E^{(2)} = \sum_{ai} \frac{|v_{ia}|^2}{\varepsilon_i - \varepsilon_a}$$

$$\left(\begin{array}{c} \varepsilon_m = \varepsilon_0 - \varepsilon_i + \varepsilon_a \\ \\ \frac{1}{\varepsilon_0 - \varepsilon_m} = \frac{1}{\varepsilon_i - \varepsilon_a} \end{array} \right)$$

$$= \sum_{ji} \frac{|\bar{v}_{ij,aj}|^2}{\varepsilon_i - \varepsilon_a}$$

2p2h $m \rightarrow a_a^\dagger a_b^\dagger a_j a_i |\Phi_0\rangle$

$\sum_{m \geq 1} \rightarrow \frac{1}{4} \sum_{\substack{a b \\ i' j'}}$

$\rightarrow \sum_{\substack{a<b \\ i'<j'}}$

$\Delta E^{(2)}(2p2h) =$

$\frac{1}{4} \sum_{\substack{a b \\ i' j'}} \frac{1}{4} \sum_{pqrs} - \frac{1}{4} \sum_{tuvw}$

$\delta_{pi} \delta_{qi'} \delta_{ra} \delta_{sb}$

$$\langle \Phi_0 | a_p^+ a_q^+ a_s a_r a_a^+ a_c^+ a_j a_i^+ | \Phi_0 \rangle$$

$$\times \langle \Phi_0 | a_j a_i^+ a_c a_a^+ a_b^+ a_d^+ a_w a_v | \Phi_0 \rangle$$

$$\frac{}{\varepsilon_0 - \varepsilon_m}$$

$$\bar{v}_{ij,ab} \qquad \bar{v}_{ab,ij}$$

$$\varepsilon_m = \varepsilon_0 - \varepsilon_{i'} - \varepsilon_j + \varepsilon_a + \varepsilon_c$$

$$= \frac{1}{4} \sum_{\substack{ab \\ ij}} \frac{|\bar{v}_{ijab}|^2}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}$$

How does this relate to FCI?

$$\Delta E = \sum_{ai} c_i^a f_{ia} + \sum_{\substack{ab \\ ij}} c_{ij}^{ab} \bar{v}_{ijab}$$

$f_{ia} = 0$

$$f_{ia} = \langle i | h_0 | a \rangle + \langle i | u | a \rangle$$

$h_0 |a\rangle = \varepsilon_a |a\rangle$

$\langle i | a \rangle = 0$

$$= \sum_j \bar{v}_{ija_j}$$

$$\Delta E = \sum_{a_i'} c_{i'}^{a} \sum_{j} \bar{v}_{i'j'a_j'}$$

$$+ \sum_{\substack{ab \\ i'j'}} c_{i'j'}^{ab} \bar{v}_{i'j'ab}$$

$$\Delta E^{(2)}(\text{RSPT}) = \sum_{\substack{a_i' \\ j}} \frac{|\bar{v}_{i'j'a_i'}|}{\varepsilon_{i'} - \varepsilon_a} + \frac{1}{4} \sum_{\substack{ab \\ i'j'}} \frac{|\bar{v}_{i'j'ab}|^2}{\varepsilon_{i'} + \varepsilon_j - \varepsilon_a - \varepsilon_b}$$

$$u_{a_i'} = \sum_j \bar{v}_{i'j a_j'}$$

$$c_{i'j'}^{ab} \simeq \frac{\bar{v}_{abi'j'}}{\varepsilon_{i'} + \varepsilon_j - \varepsilon_a - \varepsilon_b} \quad , \quad c_{i'}^{a} \simeq \frac{u_{a_i'}}{\varepsilon_{i'} - \varepsilon_a}$$